

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 34

GLACIAL LANDFORMS

- **EROSIONAL LANDFORMS**
- **DEPOSITIONAL LANDFORMS**

GLACIER TOPOGRAPHY

GLACIERS

- 1 Masses of ice moving as sheets over the land or as linear flows down the slopes of mountains (mountain and valley glaciers) are called glaciers.



- 2 It normally assumes a tongue-shape, broadest at the source becoming narrower downhill. Though the glacier is not a liquid, under the continual pressure from the accumulated snow above, it moves.
- 3 The movement of glaciers is slow unlike water flow. The rate of movement is greatest in the middle where there is little obstruction.
- 4 The sides and the bottom are held back by friction with the valley side and valley floor.

GLACIERS



5 Erosion by glacier is tremendous because of friction caused by sheer weight of the ice.

6 The material plucked from the land by glaciers (usually large sized angular blocks and fragments) get dragged along the floors or sides of the valleys and cause great damage through abrasion and plucking.

7 Glaciers can cause significant damage to even un-weathered rocks and can reduce high mountains into low hills and plains.

GLACIER TOPOGRAPHY

GLACIERS

- 8 As glaciers continue to move, debris gets removed, **divides get lowered** and eventually the **slope is reduced** to such an extent that glaciers will stop moving leaving only a mass of low hills and **vast outwash plains** along with other depositional features.



GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

1

CIRQUES



1 Cirques are the most common of landforms in glaciated mountains. The cirques quite often are found at the **heads of glacial valleys**.

2 The accumulated ice cuts these cirques while moving down the mountain tops.

3 They are deep, long, and wide troughs or basins with **very steep concave to vertically dropping high walls** at its head as well as sides.

GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

1

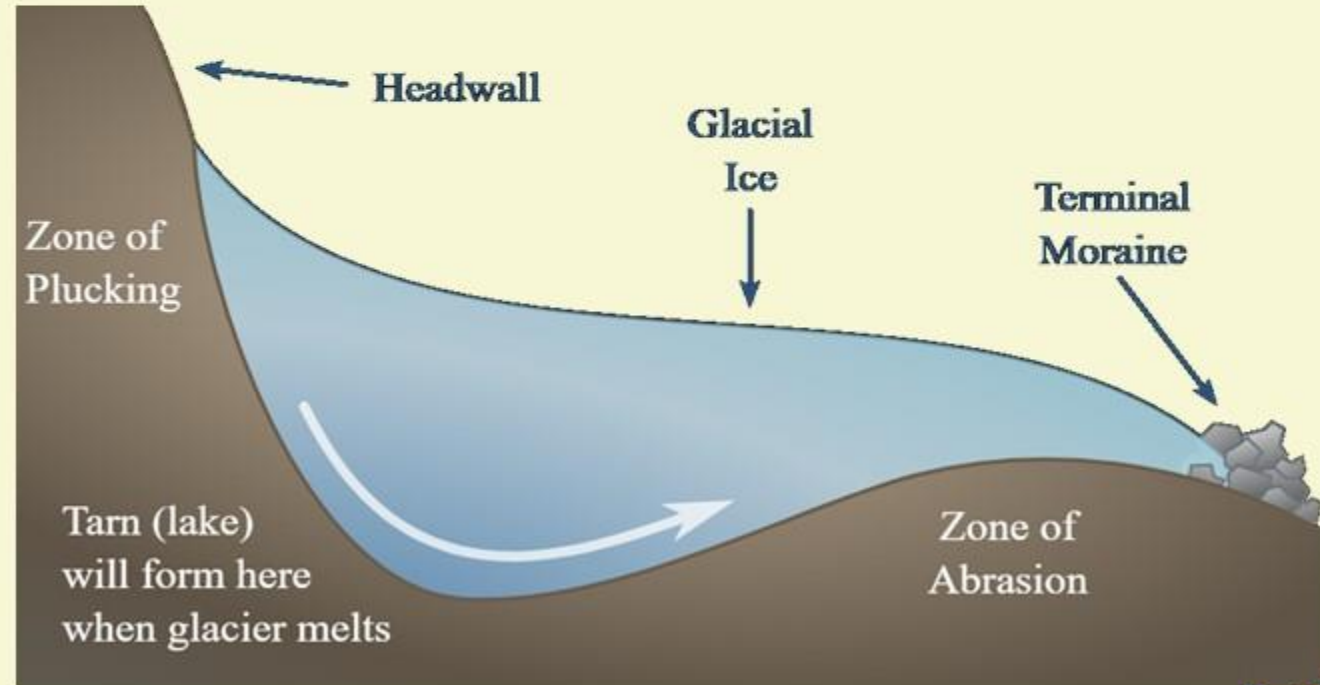
CIRQUES

4

A Lake of water can be seen quite often within the cirques after the glaciers disappears. Such lakes are called **Cirque or tarn lakes**.

5

There can be two or more cirques one leading into another down below in a stepped sequence.

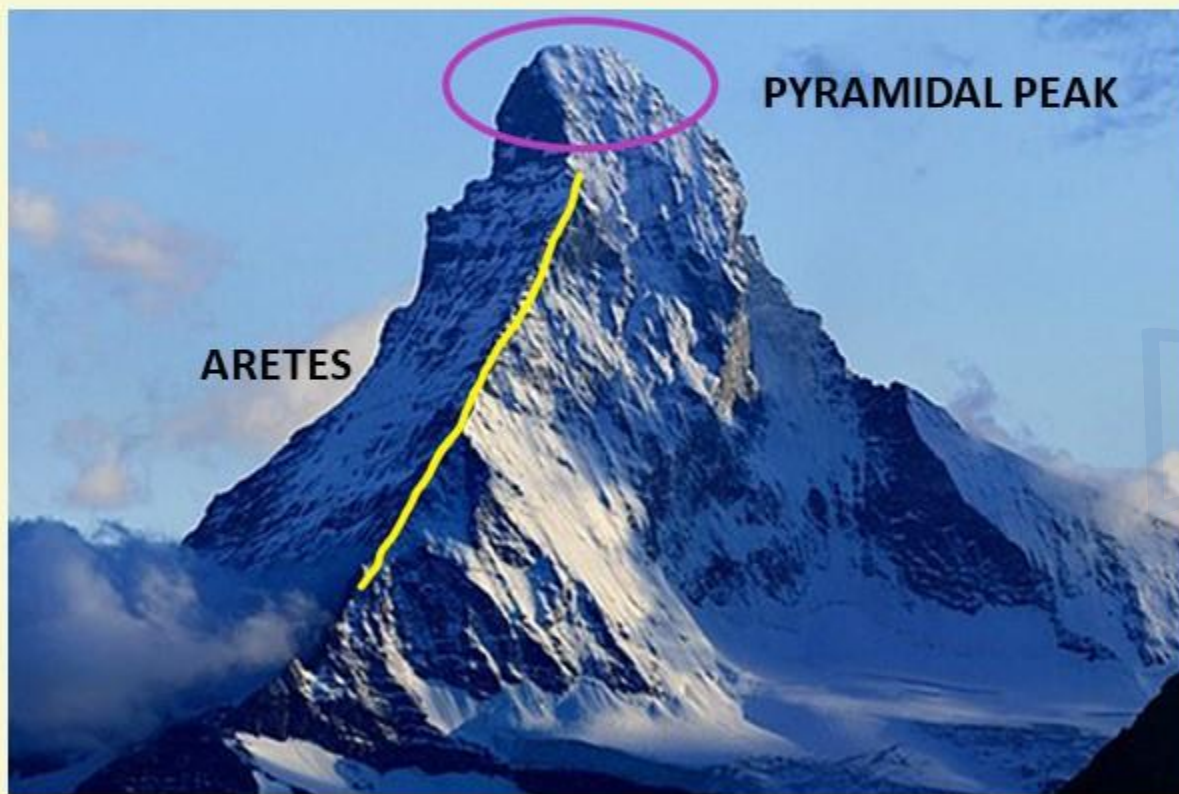


GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

2

HORNS AND SERRATED RIDGES



1

When two corries cut back on opposite sides of a mountain, **knife-edged ridges** are formed called **aretes**.

2

When three or more cut back together, their ultimate recession will form an high, sharp pointed and steep sided peaks called **angular horns** or **pyramidal peak**.

DID YOU KNOW?

The highest peak in the Himalayas, Everest are, in fact, horns formed through head ward erosion of radiating cirques.

GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

2 HORNS AND SERRATED RIDGES

- 3 The divides between cirque side walls or head walls get narrow because of progressive erosion and turn into **serrated or saw-toothed ridges** sometimes referred to as **arêtes** with very sharp crest and a zig-zag outline.



GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

3

GLACIAL VALLEYS/TROUGHS

- 1 The glaciers on its downward journey fed by ice from several corries, begins to wear away the sides and floor of the valley down which it moves.



- 2 It tends to **straighten any protruding spurs on its course**. A valley which has been so glaciated takes a characteristic **U-shape** with a wide, flat floor and very steep sides.

- 3 After the disappearance of the ice, the deep sections of these long, narrow glacial troughs may be filled with water forming **ribbon lakes**. Eg: Loch Ness lake in Britain.

- 4 There may be lakes gouged out of rocky floor or formed by debris within the valleys.

GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

3

GLACIAL VALLEYS/TROUGHS

5 There can be **hanging valleys** at an elevation on one or both sides of the main glacial valley.



6 A hanging valley is elevated above another valley, with one end open to the valley below. There may be a cliff or steep formation where they meet.

7 A river or stream may run through a hanging valley, forming a waterfall that enters the lower valley.

8 A hanging valley can be formed **when the lower valley has a greater rate of erosion**. This may be caused by a greater flow of water in the lower valley or soft rock layers that erode more quickly.

GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

3

GLACIAL VALLEYS/TROUGHS

9

Very deep glacial troughs filled with sea water and making up shorelines (in high latitudes) are called **fjords** /fiords.

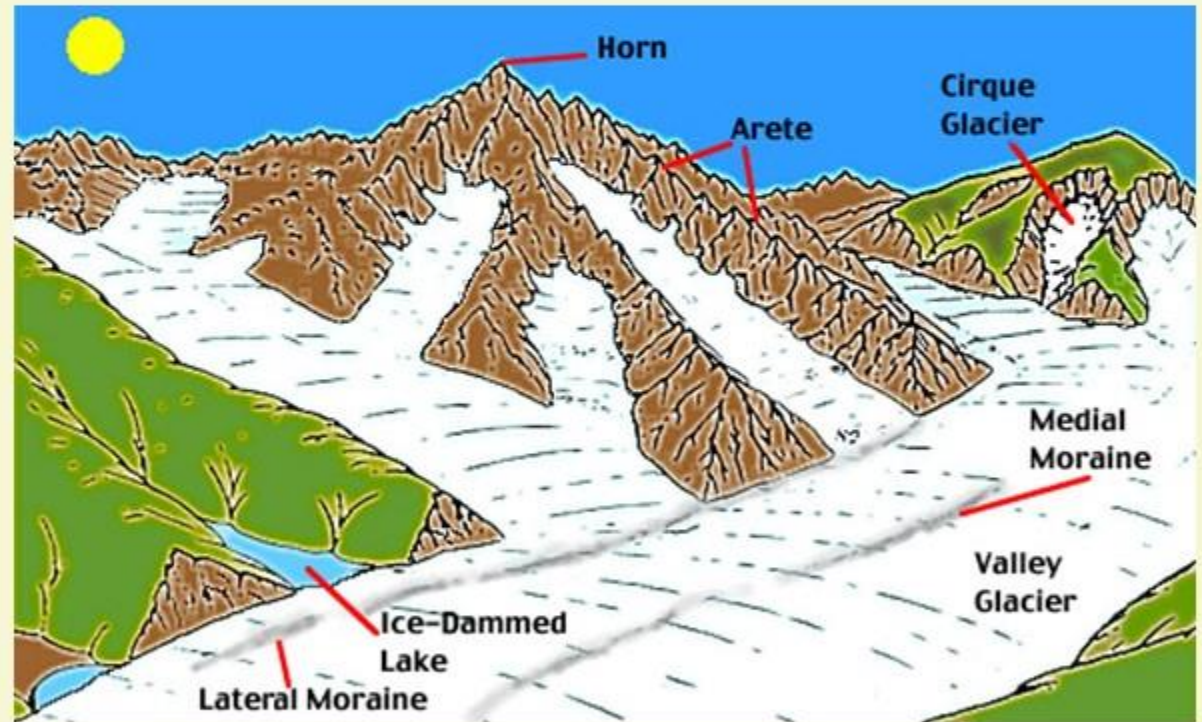
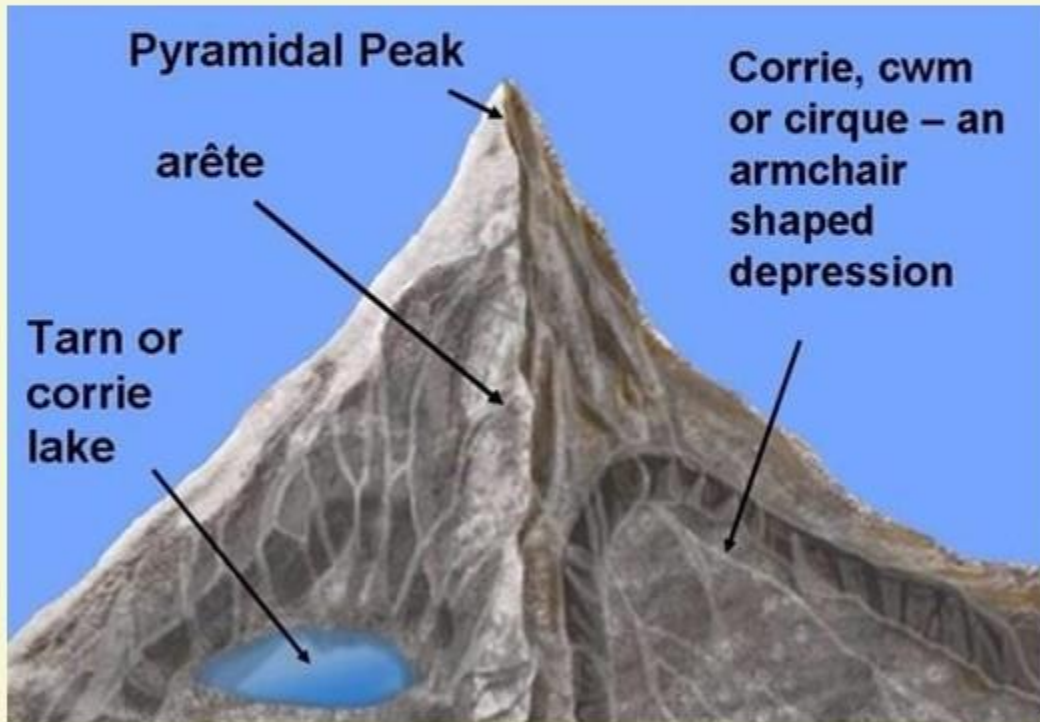


Fiords of Norway

GLACIER TOPOGRAPHY

EROSIONAL LANDFORMS

**A QUICK GLANCE AT THINGS WE
LEARNT SO FAR....**



GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS



- 1 The coarse and fine debris dropped by the melting glaciers is called **glacial till**.
- 2 Most of the rock fragments in till are **angular to sub-angular** in form.
- 3 Streams form by melting ice at the bottom, sides or lower ends of glaciers.

GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS



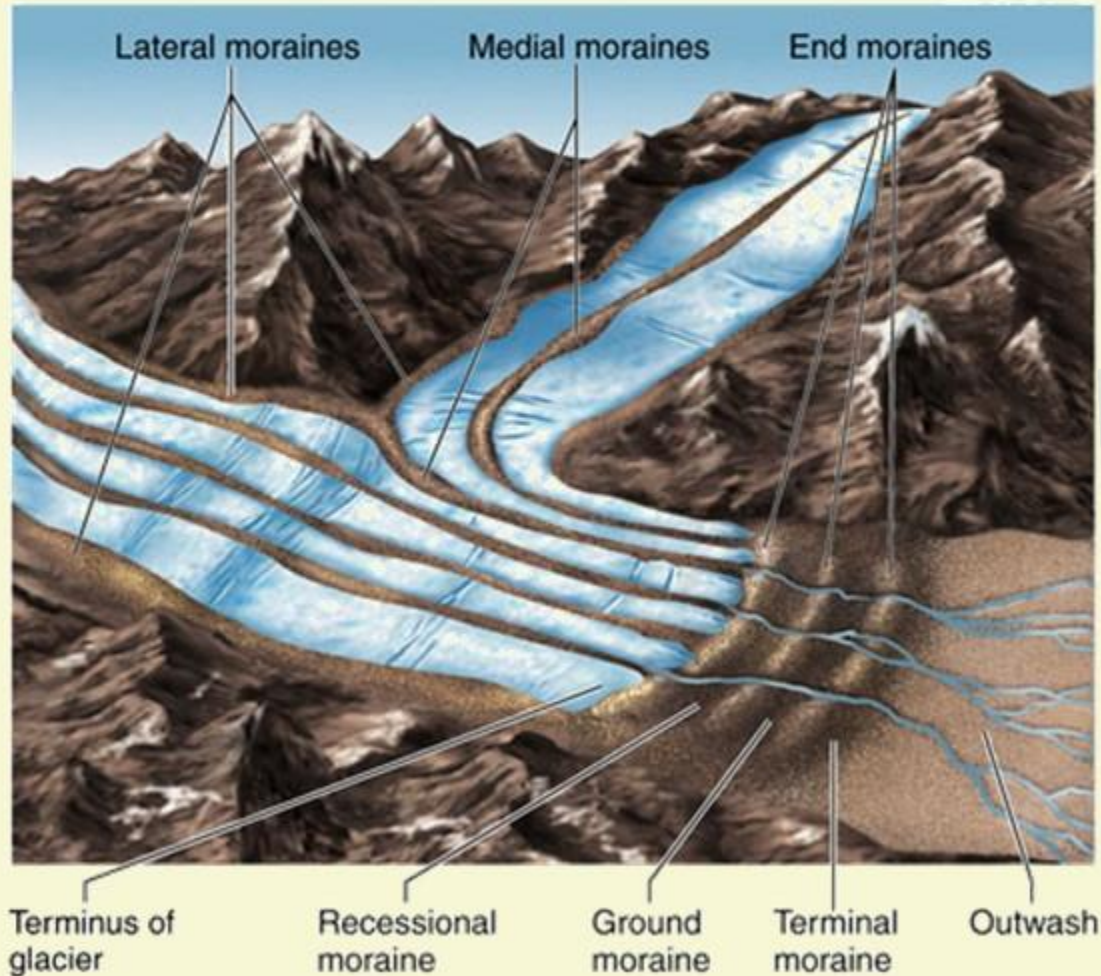
- 4 Some amount of rock debris small enough to be carried by such melt-water streams is washed down and deposited.
- 5 Such glacio-fluvial deposits are called **outwash deposits**.
- 6 Unlike till deposits, the outwash deposits are **roughly stratified and assorted**.
- 7 The rock fragments in outwash deposits are somewhat rounded at their edges.

GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

1

MORAINES



1

They are long ridges of deposits of glacial till. Terminal moraines are long ridges of debris deposited at the end (toe) of the glaciers.

2

Lateral moraines form along the sides parallel to the glacial valleys.

3

The lateral moraines may join a terminal moraine forming a horse-shoe shaped ridge.

4

There can be many lateral moraines on either side in a glacial valley. These moraines partly or fully owe their origin to glacio-fluvial waters pushing up materials to the sides of glaciers.

GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

1

MORAINES



5 Many valley glaciers retreating rapidly leave an irregular sheet of till over their valley floors.

6 Such deposits varying greatly in thickness and in surface topography are called **ground moraines**.

7 The moraine in the centre of the glacial valley flanked by lateral moraines is called **medial moraine**. They are imperfectly formed as compared to lateral moraines.

8 Sometimes medial moraines are indistinguishable from ground moraines.

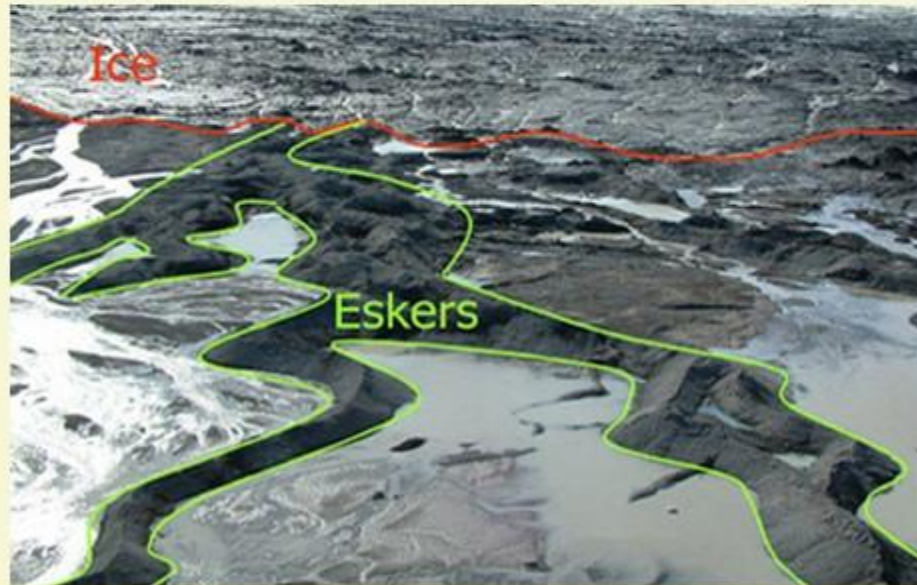
GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

2

ESKERS

- 1 When glaciers melt in summer, the water flows on the surface of the ice or seeps down along the margins or even moves through holes in the ice.



- 2 Most eskers are argued to have formed within ice-walled tunnels by streams which flowed within and under glaciers.
- 3 Eskers may also form above glaciers by accumulation of sediment in crevasses in linear zones between stagnant blocks, or at glacier margins.
- 4 Very coarse materials like boulders and blocks along with some minor fractions of rock debris carried into this stream settle in the valley of ice and after the ice melts can be found as a sinuous ridge called esker.

GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

3

OUTWASH PLAINS

1

The plains at the foot of the glacial mountains are covered with glacio-fluvial deposits which are washed out of terminal moraines and spread in the form of broad flat fan-like structures called outwash plains.



GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

3

OUTWASH PLAINS

2

Where the deposition in these outwash plains takes the form of alternating ridges and depressions, the depressions may get filled with melt-water and form 'kettle lakes'. This gives rise to a characteristic 'knob and kettle' topography.



GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

4

DRUMLINS

1 Drumlins are smooth **oval shaped ridge-like features** composed mainly of glacial till with some masses of gravel and sand.

2 The long axes of drumlins or their elongated sides are **parallel to the direction of ice movement**.



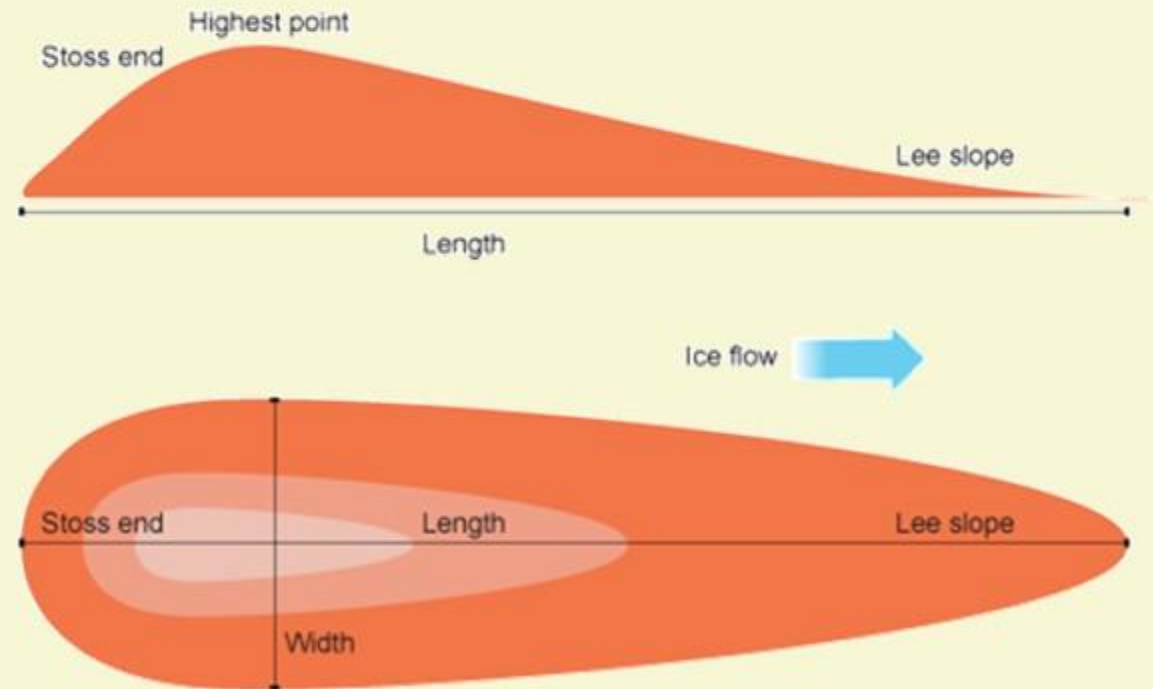
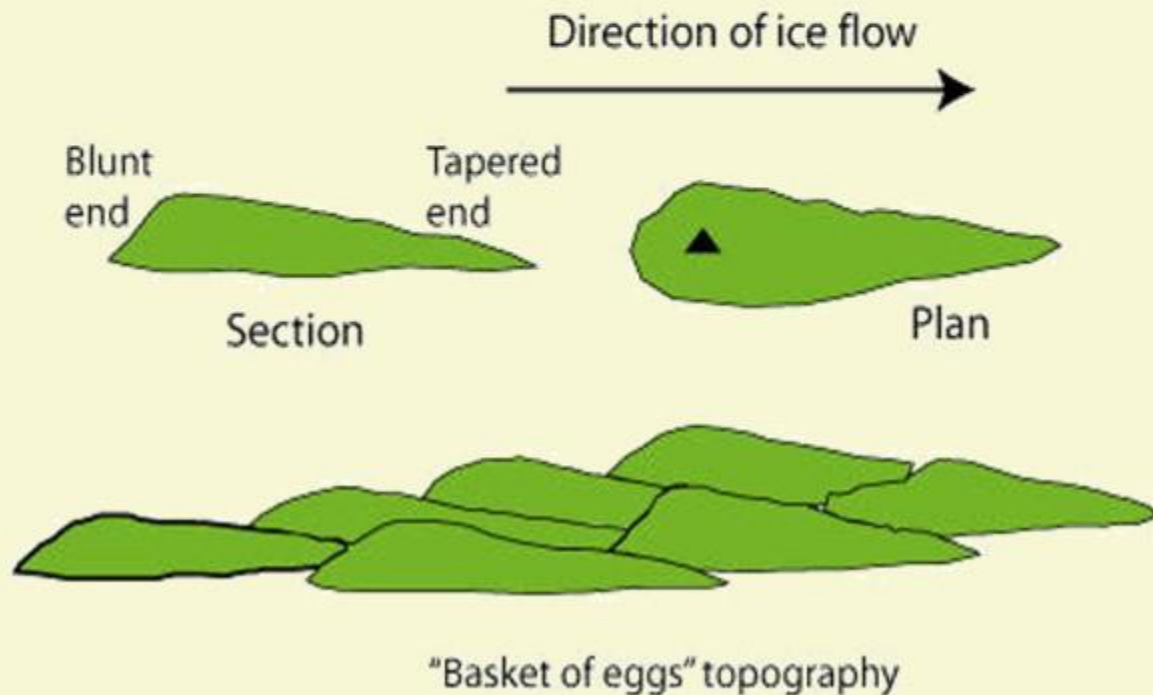
GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

4

DRUMLINS

- 3 They may measure up to 1 km in length and 30 m or so in height. They appear a little steeper at the onset side (ie the side facing the glacier) and this is called the **stoss end**. The drumlins taper off at the other leeward end called the **tail**.



GLACIER TOPOGRAPHY

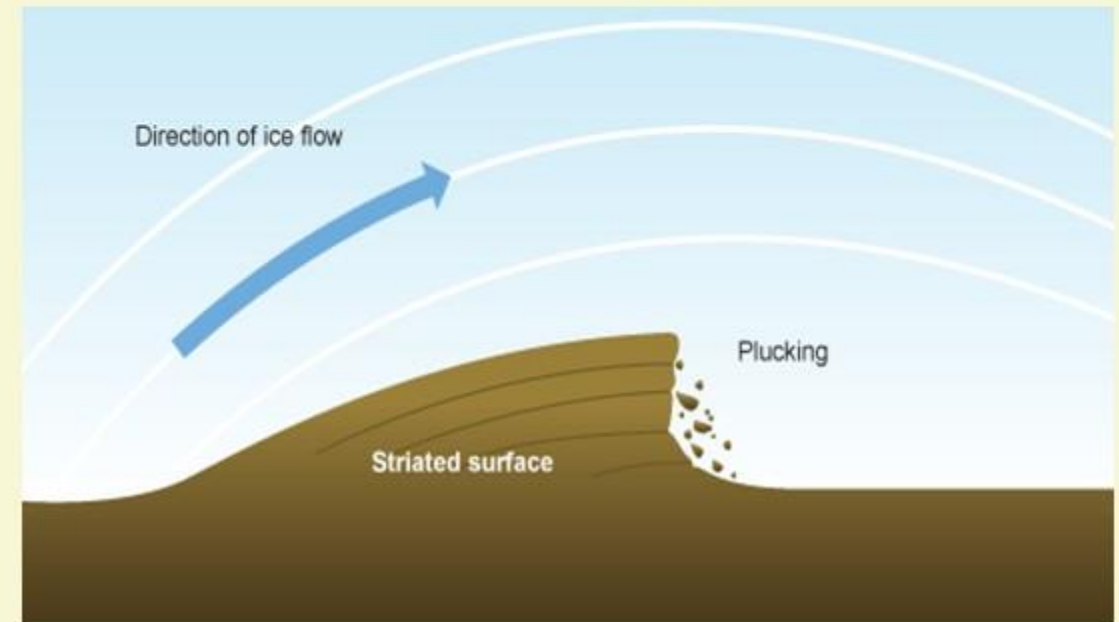
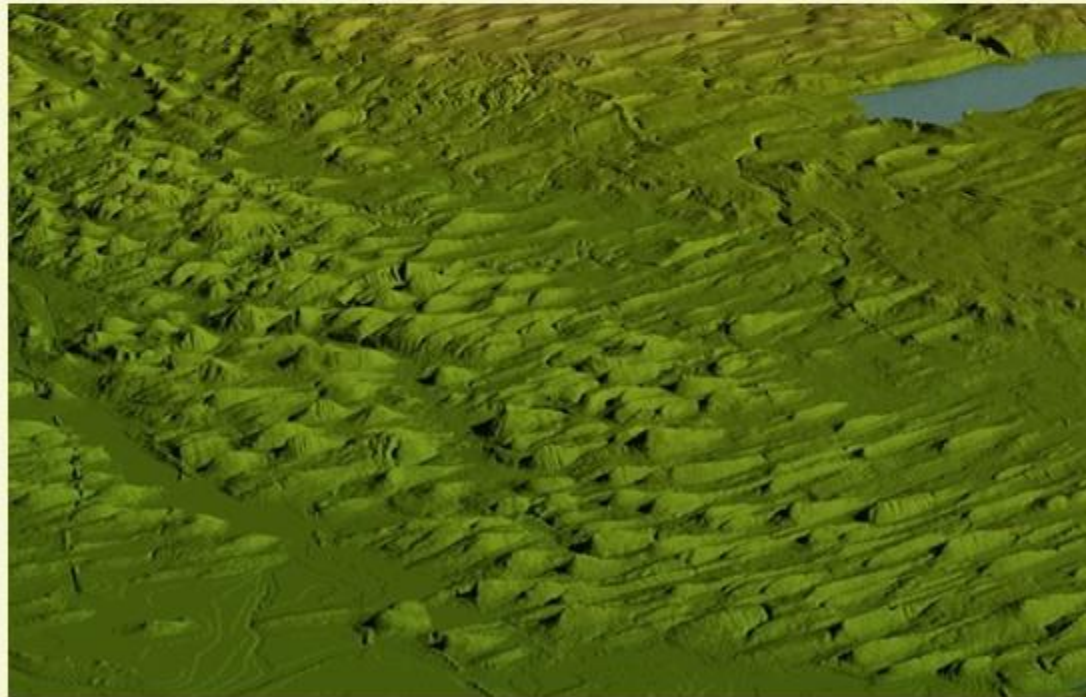
DEPOSITIONAL LANDFORMS

4

DRUMLINS

4 The drumlins form due to dumping of rock debris beneath heavily loaded ice through fissures in the glacier.

5 Drumlins give an indication of direction of glacier movement. The stoss end gets blunted due to pushing by moving ice.



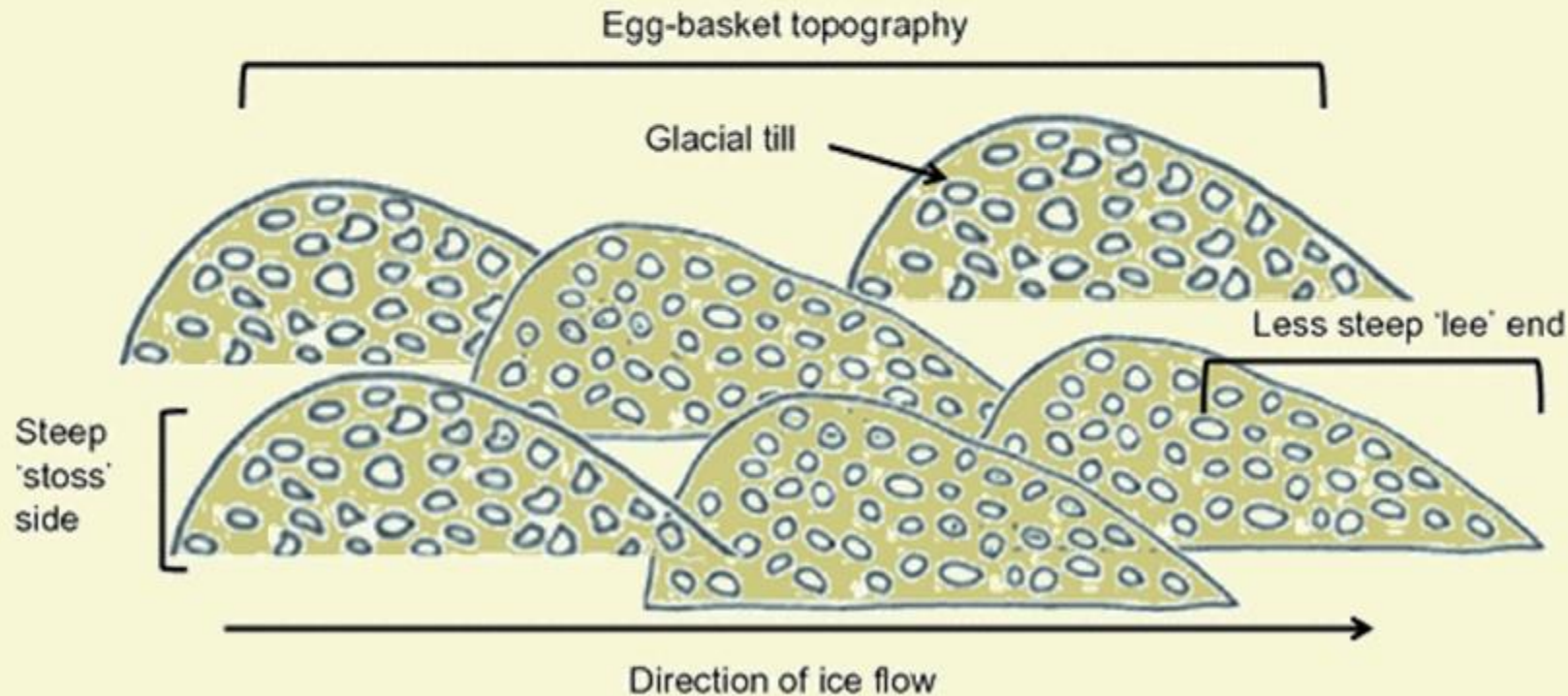
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DEPOSITIONAL LANDFORMS

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DRUMLINS

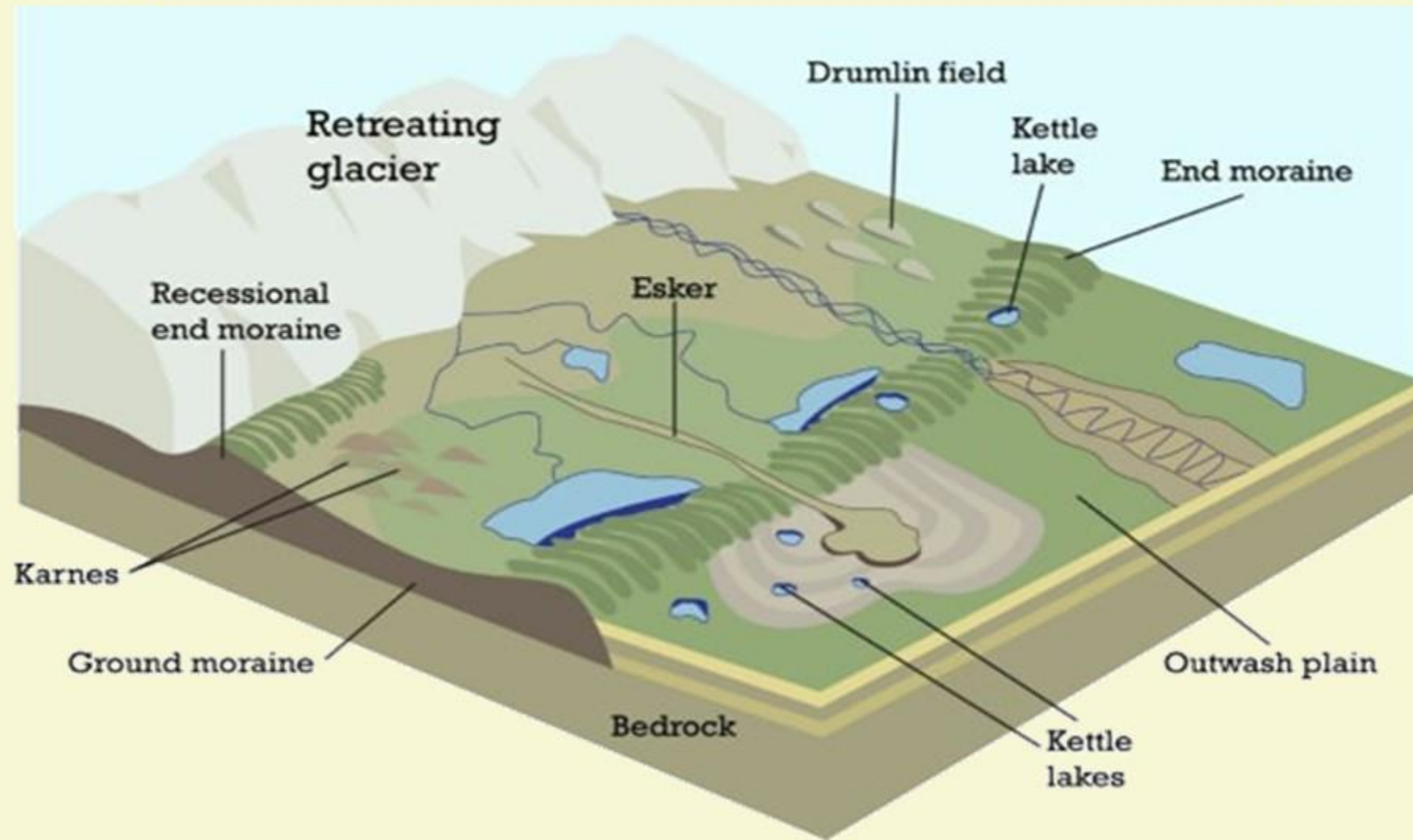
- 6 They are arranged diagonally and so are commonly described as having a 'basket of eggs' topography.
E.g.: the glaciated plain around the Great Lakes in North America.



GLACIER TOPOGRAPHY

DEPOSITIONAL LANDFORMS

A QUICK GLANCE AT THINGS WE LEARNT SO FAR....



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